

Report generated on

PVGIS-5.3 geo-temporal irradiation database

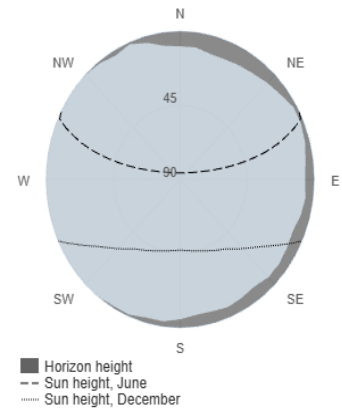
Provided inputs

Latitude/Longitude: 19.456,72.828
Horizon: Calculated
Database used: PVGIS-ERA5
Start year: 2005
End year: 2023

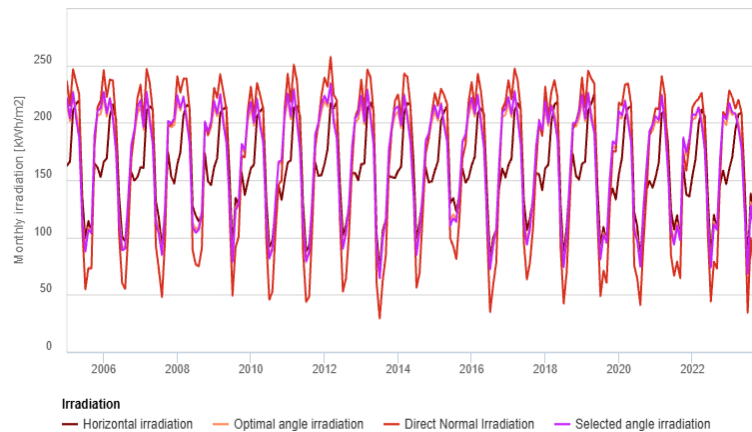
Variables included in this report:

Global horizontal irradiation: Yes
Direct Normal Irradiation: Yes
Global irradiation optimum angle: Yes
Global irradiation at angle 30°: Yes
Diffuse/global ratio: Yes
Average temperature: Yes

Outline of horizon at chosen location:



Monthly solar irradiation estimates



Global horizontal irradiation

Month	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
January	162.44	165.84	161.06	163.82	160.94	160.21	165.01	163.81	163.88	157.92	158.24	162.83	163.12	160.4	164.15	154.96	152.67	153.77	159.59
February	166.11	169.12	160.72	174.14	169.22	163.54	167.32	176.79	164.16	161.53	166.37	170.17	168.87	163.52	166.08	169.07	165	168.37	170.32
March	210.67	205.87	210.42	206.68	208.76	205.31	212.69	217.14	212.08	208.36	201.69	208.38	211.03	200.3	210.23	203.84	208.29	198.54	193.95
April	216.51	215.94	214.14	215.37	211.96	209.88	214.61	212.39	217.37	217.06	209.35	212.2	216.93	216.11	216.37	213.01	210.46	211.41	207.1
May	219.39	203.49	211.59	213.62	213.6	213.57	205.67	218.55	210.96	215.48	214.85	207.95	211.42	217.62	225.09	214.07	182.55	205.81	209.12
June	134.97	136.53	131.9	127.24	156.3	134.26	128.65	147.97	102.06	166.98	129.94	137.32	132.33	128.39	143.69	122.19	123.94	136.81	147.67
July	99.16	101.01	116.45	119.09	88.49	91.51	88.24	101.57	71.37	96.3	134.2	80.36	106.47	82.95	91.14	109.66	106.96	82.64	76.25
August	114.26	96.35	90.62	113.99	134.45	97.06	93.27	110.97	106.02	114.06	122.48	99.49	116.71	110.6	109.28	79.87	119.4	119.66	138.12
September	103.48	132.56	125.33	119.94	127.82	128.53	130.3	126.9	115.92	133.9	138.13	107.01	133.86	149.72	95.25	124.32	97.56	106.41	105.97
October	164.45	156.63	174.08	173.09	156.64	146.19	165.18	156.09	153.89	160.87	160.11	144.27	156.08	172.45	133.56	141.54	162.4	146.69	165.39
November	160.8	149.49	153.26	148.92	136.93	132.54	153.75	156.14	152.55	147.87	147.82	160.15	154.94	153.85	142.56	149.05	137.41	158.2	148.59
December	152.64	152.65	146.98	145.68	149.18	150.18	154.26	149.89	151.86	148.92	151.78	152.19	141.1	150.43	133.01	143.36	135.81	146.46	144.14

Direct Normal Irradiation

Month	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
January	235.65	245.77	233.14	240.53	230.31	231.4	242.61	239.32	237.5	224.57	225.95	235.36	236.97	231.43	239.44	217.52	212.49	213.39	228.22
February	214.91	222.33	201.85	226.37	221.67	205.86	218.11	231.54	210.51	204.63	214.79	212.58	220.39	204.94	213.69	215.14	211.82	218.15	222.11
March	246.53	237.72	246.82	238.21	242.16	234.68	250.56	257.34	246.39	242.71	229.35	242.69	247.18	226.94	245.29	233.22	240.58	220.87	212.47
April	235.34	236.66	234.88	238.51	226.93	224.04	236.77	224.41	239.59	240.13	224.16	227.68	236.9	237.12	238.6	234.04	224.06	226.12	219.82
May	224.77	199.99	209.74	211.84	211.5	212.46	199.23	219.57	208.8	215.58	215.98	203.33	211.29	222.15	234.18	215.95	167.65	198.35	208.8
June	109.25	103.24	93.01	88.95	124.95	98.85	88	116.87	59.82	143.69	99.11	107.64	96.29	91.17	116.69	75.68	84.33	103.72	131.28
July	54.91	60.09	71.52	76.28	49.25	45.74	43.82	52.86	29.44	56.22	92.27	34.99	63.21	42.46	48.7	64.05	66.47	44.23	34.52
August	73.16	55.24	48.06	74.95	91.49	52.55	48.25	64.35	62.88	68.63	81.06	58.41	77.9	70.09	71.02	41.05	78.83	78.93	99.48
September	72.95	103.49	97.64	90.59	100.43	104.03	107.82	103.66	84.16	118.2	116.78	77.74	107.79	131.06	60.35	94.58	64.52	72.97	72.82
October	179.07	168.91	196.34	197.97	171.15	145.87	179.38	162.6	160.2	172.8	168.88	153.1	161.99	194.64	124.76	136.83	174.74	150.76	183.25
November	213.22	191.32	198.22	189.53	169.73	149.69	199.84	204.83	196.34	186.17	181.01	211.9	202.54	200.21	175.16	188.57	158.04	209.3	188.58
December	219.34	221.44	203.79	199.55	210.66	212.42	223.62	211.97	219.13	208.32	218.61	218.08	188.39	210.68	174.95	199.21	180.9	205.42	197.38

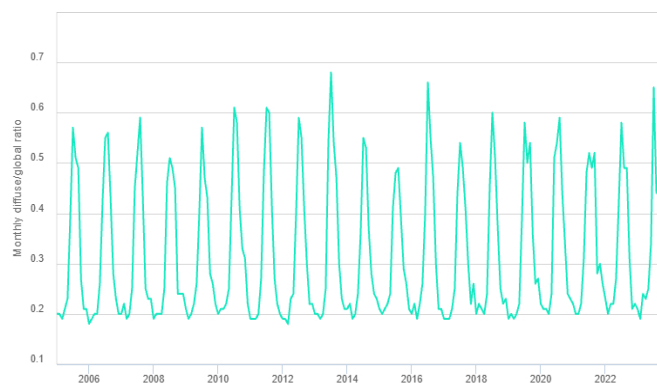
Global irradiation optimum angle

Month	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
January	216.4	221.72	214.59	218.72	213.85	213.09	220.31	218.54	218.38	209.55	210.16	216.71	217.38	213.35	219.01	205.13	201.26	202.86	212.09
February	201.41	205.52	193.95	210.52	205.48	197.44	203.03	214.21	198.48	195.27	201.66	205.23	204.86	197.15	201.07	203.92	199.86	204.12	206.59
March	227.19	221.87	227.12	223.18	225.35	221.27	229.61	234.58	229.02	224.75	217.29	224.72	227.96	215.58	226.66	219.69	224.62	213.73	208.5
April	208.06	207.6	205.79	206.62	203.49	201.48	206.29	204.25	208.86	208.49	201.41	203.98	208.38	207.48	207.91	204.67	202.5	203.32	199.15
May	193.23	180.65	187.21	188.98	189.09	189	182.29	193.04	186.83	190.45	189.91	184.4	186.96	191.98	197.92	188.93	163.16	182.72	184.55
June	117.59	119.84	116.63	112.5	136.17	118.34	113.86	129.32	91.68	144.34	114.1	120.12	116.69	113.46	125.16	109.25	109.9	120.04	127.25
July	90.07	91.44	105.54	107.18	80.73	83.7	80.68	92.52	65.87	87.18	119.85	73.91	96.36	76.04	83.13	99.62	96.34	75.71	69.98
August	108.72	91.89	86.1	108.68	126.47	91.73	88.43	105.34	100.7	107.75	115.9	94.49	109.62	104.32	104.15	75.93	112.67	113.58	130.56
September	104.84	135.12	127.13	122.83	130.63	131.91	133.91	130.29	116.96	138.21	141.18	107.69	136.32	153.28	96.73	126.06	98.37	107.53	106.85
October	188.48	179.67	200.24	199.74	180.34	165.02	189.36	178.54	175.74	183.16	182.84	165.88	178.16	198.37	149.17	160.14	186.17	167.55	190.08
November	206.13	190.27	195.87	188.74	173.24	164.79	196.11	199.95	194.56	187.58	186.56	205.24	198	196.72	180.51	189.29	171.51	202.72	188.75
December	207.44	207.9	198.28	195.88	202.08	203.59	210.14	203.03	206.63	201.25	206.41	206.79	188.86	203.48	177.09	193.22	181.61	197.95	193.87

Global irradiation at angle

Month	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
January	221.47	226.99	219.63	223.9	218.83	218.05	225.52	223.7	223.51	214.39	215.04	221.78	222.48	218.32	224.18	209.83	205.78	207.44	217.03
February	203.85	208.05	196.22	212.98	207.99	199.75	205.51	216.77	200.83	197.6	204.09	207.61	207.34	199.43	203.47	206.27	202.26	206.58	209.09
March	226.34	221.05	226.3	222.42	224.55	220.47	228.78	233.74	228.22	223.93	216.51	223.9	227.16	214.76	225.81	218.9	223.81	212.96	207.74
April	203.72	203.28	201.48	202.23	199.25	197.27	201.97	200.05	204.47	204.1	197.26	199.76	204.01	203.1	203.55	200.39	198.35	199.12	195.06
May	186.45	174.61	180.84	182.54	182.68	182.56	176.17	186.39	180.5	183.91	183.39	178.21	180.57	185.3	190.88	182.39	157.99	176.65	178.17
June	113.47	115.83	112.91	108.92	131.41	114.47	110.28	124.87	89.08	139.04	110.28	116	112.9	109.83	120.76	106.04	106.48	116.01	122.46
July	87.72	88.98	102.69	104.15	78.67	81.62	78.67	90.15	64.37	84.83	116.31	72.18	93.75	74.17	81.01	96.99	93.64	73.83	68.31
August	106.61	90.19	84.48	106.6	123.75	89.93	86.74	103.3	98.76	105.56	113.55	92.68	107.25	102.17	102.16	74.51	110.32	111.31	127.82
September	103.8	133.82	125.82	121.77	129.43	130.76	132.77	129.17	115.71	137.07	139.84	106.51	134.98	151.83	95.85	124.76	97.36	106.43	105.71
October	189.51	180.69	201.38	200.96	181.44	165.67	190.4	179.5	176.66	184.01	183.78	166.9	179.07	199.51	149.58	160.84	187.2	168.45	191.18
November	210.06	193.77	199.56	192.11	176.34	167.46	199.75	203.75	198.18	190.97	189.85	209.14	201.71	200.43	183.76	192.73	174.35	206.57	192.19
December	212.8	213.31	203.27	200.75	207.25	208.8	215.61	208.21	211.99	206.34	211.76	212.13	193.49	208.65	181.34	198.06	186.04	202.96	198.69

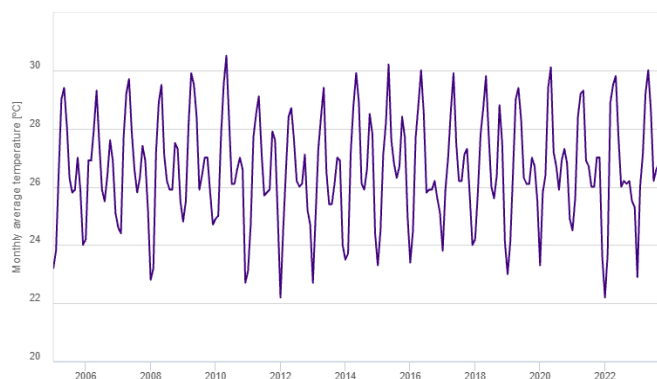
Monthly average diffuse to global ratio



Diffuse/global ratio

Month	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
January	0.2	0.18	0.2	0.19	0.21	0.2	0.19	0.19	0.2	0.21	0.21	0.2	0.19	0.2	0.19	0.22	0.22	0.23	0.21
February	0.2	0.19	0.22	0.2	0.19	0.21	0.19	0.19	0.2	0.22	0.2	0.22	0.19	0.22	0.2	0.21	0.2	0.2	0.19
March	0.19	0.2	0.19	0.2	0.2	0.21	0.19	0.18	0.19	0.19	0.21	0.19	0.19	0.21	0.19	0.21	0.2	0.22	0.24
April	0.21	0.2	0.2	0.2	0.22	0.22	0.2	0.23	0.2	0.2	0.22	0.22	0.21	0.2	0.2	0.2	0.22	0.22	0.23
May	0.23	0.26	0.25	0.25	0.26	0.25	0.27	0.24	0.25	0.24	0.24	0.26	0.25	0.24	0.22	0.24	0.31	0.27	0.25
June	0.39	0.42	0.45	0.46	0.4	0.43	0.48	0.4	0.54	0.36	0.41	0.4	0.44	0.45	0.39	0.51	0.48	0.42	0.34
July	0.57	0.55	0.52	0.51	0.57	0.61	0.61	0.59	0.68	0.55	0.48	0.66	0.54	0.6	0.58	0.54	0.52	0.58	0.65
August	0.51	0.56	0.59	0.49	0.47	0.58	0.6	0.55	0.54	0.53	0.49	0.55	0.49	0.51	0.5	0.59	0.49	0.49	0.44
September	0.49	0.43	0.43	0.45	0.43	0.41	0.41	0.41	0.47	0.37	0.39	0.47	0.41	0.37	0.54	0.44	0.52	0.49	0.49
October	0.27	0.28	0.25	0.24	0.28	0.33	0.27	0.3	0.3	0.28	0.29	0.3	0.3	0.25	0.36	0.34	0.28	0.31	0.27
November	0.21	0.23	0.23	0.24	0.26	0.31	0.22	0.22	0.23	0.24	0.26	0.21	0.22	0.22	0.26	0.24	0.3	0.21	0.24
December	0.21	0.2	0.23	0.24	0.22	0.22	0.2	0.22	0.21	0.23	0.21	0.21	0.26	0.23	0.27	0.23	0.26	0.22	0.24

Monthly average temperature



Monthly average temperature

Month	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
January	23.2	24.2	24.6	22.8	24.8	24.9	23.1	22.2	22.7	23.5	23.3	23.4	23.8	24.2	23	23.3	24.5	22.2	22.9
February	23.8	26.9	24.4	23.2	25.5	25	24.7	24.5	25	23.7	24.5	24.5	25.9	25.8	24.1	25.8	25.6	23.7	26
March	26.5	26.9	27.7	27.2	28.1	27.7	27.7	26.5	27.3	27.2	27.1	27.7	27	27.8	26.5	26.4	28.4	28.9	27.1
April	29	28	29.2	28.9	29.9	29.5	28.5	28.4	28.4	28.8	28.2	28.8	28.6	28.7	29	29.4	29.2	29.5	29.2
May	29.4	29.3	29.7	29.5	29.5	30.5	29.1	28.7	29.4	29.9	30.2	30	29.9	29.8	29.4	30.1	29.3	29.8	30
June	28.1	27.5	27.9	27.1	28.4	28.3	27.2	27.6	26.5	28.9	27.6	28.5	27.5	27.8	28.3	27.2	26.9	27.7	28.6
July	26.3	25.9	26.6	26.2	25.9	26.1	25.7	26.2	25.4	26.1	26.8	25.8	26.2	26	26.3	26.7	26.7	26	26.2
August	25.8	25.5	25.8	25.9	26.4	26.1	25.8	26	25.4	25.9	26.3	25.9	26.2	25.6	26.1	25.9	26	26.2	26.6
September	25.9	26.4	26.3	25.9	27	26.6	25.9	26.1	26.1	26.6	26.7	25.9	27.1	26.4	26.1	26.9	26	26.1	26.8
October	27	27.6	27.4	27.5	27	27	27.9	27.1	27	28.5	28.4	26.2	27.3	28.8	27	27.3	27	26.2	28
November	25.9	26.9	26.9	27.3	25.8	26.6	27.6	25.2	26.9	27.8	27.7	25.6	25.7	27.5	26.7	26.8	27	25.5	27.4
December	24	25.1	25.2	25.5	24.7	22.7	25	24.7	24	24.4	25	25.1	24	24.1	25.5	24.9	23.6	25.3	25.1